

CUSTOMER MILESTONE BRIEF

MS #69 - iDempiere Deterministic Semantic Comparison

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete bounded baseline with open semantic findings
Release	0.69.0
Date	2026-09-11
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #69 ran all 1,078 paired migrations through a deterministic comparator. It decided 776 of 111,293 SQL units (0.70 percent). The remaining uncertainty is recorded by source range and reason; this baseline does not establish migration equivalence.

Why this matters to a customer

Measures how much of the paired migration SQL a deterministic comparator can actually decide and makes the unresolved surface explicit before any model spend.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- A quote-aware lexer and bounded projection of schema changes and DML structure, retaining unsupported forms.
- Oracle MODIFY and PostgreSQL t_alter_column expansion using the pinned helper definition, with dynamic catalog and dependent-view behavior unresolved.
- A sealed 93-pair order-to-cash pilot followed by 985 remaining pairs, preserving operation order and repeated writes.
- Coverage of 111,293 SQL units: 776 parsed and compared, 103,922 parsed but indeterminate, and 6,595 unparsed. Another 4,127 administrative units are separately excluded.
- One pair equivalent under the static policy, zero proven declared-effect divergences, and 1,077 indeterminate pairs.
- Source-range findings, pilot and full reports, an unsigned content-addressed receipt, schema, cross-platform commands, source replay and adversarial tests.

How it differs from earlier work

MS #68 supplied the exact source pin, pairing manifest and order-to-cash pilot. MS #69 reuses those artifacts and the MS #33 semantic vocabulary. MS #70 will adapt the existing Planner and Analyst workcell to bounded findings; no model workcell is rebuilt or called here.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use idempotent deterministic semantic comparison as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- The completed baseline has low decision coverage. DML needs baseline column, trigger, predicate and coercion evidence; it is not certified equivalent.
- Coverage counts statements and opaque blocks on both dialect sides, not lexical occurrences. Malformed input can form one opaque remainder unit.
- Equivalence covers ordered declared schema effects only. It excludes final database state, successful execution, registration/client effects and runtime behavior.
- Character, datetime, unconstrained numeric, catalog and helper-deployment obligations remain unresolved. Maintenance provenance is unclassified.
- No agents were called or work orders created. MS #70 must bound and budget a sample before using the existing workcell.
- No native execution, application equivalence, audit completion, migration completion or production readiness is claimed. Earlier signed CloudBank evidence is unchanged.

Source of record

- CHANGELOG.md release section(s): 0.69.0
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity